

# Labeling and head-movement in the left periphery: the case of Old Irish verbal complexes

Phil Branigan and Alain Rouveret

## Abstract

The arrangement of words in Old Irish clauses was rigidly VSO. Although the option also existed to place a topicalized or focalized phrase in initial position, the sentence usually began with a verbal complex which could include a preverbal particle, one or several preverbs and an inflected finite verbal form, which typically worked together to establish the core lexical semantics of the verb phrase. The verbal complex was itself followed by a subject noun phrase, an object phrase in transitive clauses, adverbials and prepositional phrases. Old Irish verbal clusters display an intricate set of syntactic, morphological, phonological, properties, which, at first sight, appear to be unrelated. It is shown that a unified account of these patterns can be provided, building on the cartographic approach and resorting to labeling theory to motivate the various head movements that take place. The analysis offers new insights into the role of feature strength in labeling endocentric structures and in justifying the deletion of phase heads.

## 1 Introduction

Old Irish verbal complexes are grammatical objects which display an unusually intricate cluster of syntactic, morphological and phonological properties. The question arises as to whether it is possible to provide a unitary syntactic account of these properties or whether specific references must be made to the other levels of grammatical organisation, namely morphology and phonology. We propose a mixed analysis of Old Irish verbal complexes in which most of their salient characteristics are accounted for on a strictly syntactic basis, but where limited reference to specific mechanisms operating after syntax turns out to be necessary, as some previous analyses ([Adger 2006](#), [2000](#), [Newton 2007](#), [Newton 2006](#)) have argued (the reason for this is different from the one they invoke). In fact, most of the grammatical information necessary to generate the right sequences is present at the syntactic level; the fact that sub-word constituency is clearly involved in the phenomenon raises no difficulty against this conclusion in a model in which syntax subsumes morphology. It must be acknowledged that the facts under consideration are

extremely complex and that the solutions proposed to account for them cannot avoid a degree of commensurate complexity.

It will be convenient to proceed in two steps and to successively consider (i) the internal categorial make-up of verbal clusters, in order to determine the nature of the functional sequence that underlies them and the position each sub-constituent occupies, and (ii) the derivational path that is followed by each element to reach its Spell-Out position. The following assumptions, which respectively address the representational and derivational aspects of the phenomenon, will be introduced and justified.

- (1) The general form verbal complexes take reflects the strategy selected by Old Irish to satisfy the labeling requirements of the left periphery.
- (2) The derivation of Old Irish verbal complexes necessitates multiple head movement to the periphery.

The first claim is significant for our understanding of the combinations of formal features and root information that provide a label acceptable to interface interpretation and, more generally, of the intricacies of the labeling algorithm. The second proposal provides new evidence for a model of head movement that can account for the data under consideration and for a type of head movement operation—multiple head movement—which has been documented only minimally up to this point.

## **2 The functional sequence in Old Irish verbal clusters and the labeling requirement**

In recent contributions, (Chomsky 2008, 2013, 2015) develops the idea that complex syntactic objects need a label in order to be interpretable at the CI interface and to function as the inputs to the process of externalization. The necessity to label certainly holds for internal syntactic projections, but it takes a special relevance when the left edge is considered because, in this case, it is the label of the whole sentential domain that is at stake. In Chomsky's system, labeling is separated from the principles that govern the construction of expressions. Syntactic structures don't include pre-labeled positions, since positions in the bare phrase structure theory he adopts don't exist independently of the content they supposedly host. Two cases must be considered. When the highest element in the clausal architecture is a head, whether simple or complex, it is the feature on this head that provides the whole clause with a label. When the highest element is a phrase, a feature-sharing "agreement" relation must exist between this phrase and the projection it is adjoined to. This claim is intended to solve a labeling difficulty, stemming from the fact that the configurations associating two projections standing in a symmetric relation are unlabelable, since they contain two heads that can potentially provide a label. We will adopt the following formulation of the Labeling Algorithm (cf. Chomsky (2013)).

- (3) *Labeling Algorithm*

- a. In [H, XP] structures, H a head, the feature(s) of H provides the label.
- b. In [XP, YP] structures, If X and Y are identical in one respect and, in particular, share some feature, this feature is selected as the label of SO; in this case, the label is the pair of agreeing elements.

There is no doubt that all well-formed Old Irish sentences, including those that begin with a verbal cluster, fulfill the labeling requirement. The aim of this section is to identify the label that is selected by the various constructions under study.

## 2.1 Some salient properties of Old Irish (OI) verbal clusters

In its most extended form, the OI verbal complex in declarative clauses could include a conjunct particle (sometimes referred to as "preverbal particle"), an infixed object pronoun, one or several lexical preverbs, a verbal stem and an inflectional ending, strictly in this order:<sup>1</sup>

- (4) particle < infixed pronoun < preverb1 < preverb2 < ... < verbal stem < inflection

All these elements are represented in the following example.<sup>2</sup>

- (5) *ní=m a-ccai.* [ad cí 'sees']  
 NEG=1s PV-see.PRS.3s  
 'He doesn't see me.'

But not all the formatives listed in (4) are morphologically represented in all verbal complexes. The identity of the first element in the sequence may vary. It is the negation, hence a preverbal particle, in (6a), a preverb in (6b). It can also be a complementiser like *co/con* "so that", *día* "if", *ara* "in order that", as also shown in (6b).

- (6) a. *ní beir in fer in claideb.*  
 NEG carry.PRS.3s the man the sword  
 'The man doesn't carry the sword.' (MI 102<sup>d</sup>3; [Newton \(2006: 1\)](#))
- b. *as ru-bart día friu-som ara celebratis a sollumnu.*  
 PVPRF-carry.3s god to.them that celebrate.3p.SUBJ his feasts  
 'God has said to them that they should celebrate his feasts.'

1. Most of the OI examples used in this article are borrowed from [Thurneysen \(1946\)](#), [Strachan \(1909\)](#), [Watkins \(1963\)](#), and [McCone \(1997\)](#). Some of them are quoted and commented in [Carnie, Harley & Pyatt \(2000\)](#), [Adger \(2006\)](#), [Newton \(2006\)](#), [Newton \(2007\)](#). We have tried to provide each example with a precise reference. It is either the original source, when we know it, and the page of the handbook where it is quoted, or the linguistic article where it is used.

2. Although the presentation of the OI data in a fully transparent format would require that some unpacking of the morphophonology be systematically provided (it has occasionally been done), the source data will mostly be presented unaltered. The original data and glosses basically follow Leipzig conventions. We depart from these conventions in two ways. First, object clitics are separated from their supporting host by the = sign only in the glosses, not in the original examples. Second, in the presentation of *deuterotonic* verbal complexes, in which the first constituent is separated from the rest by a *juncture* which motivates much of the analysis. In our presentation, this juncture is represented with a space, departing from the raised dot convention commonly adopted in the philological literature on OI.

The verbal cluster may also reduce to a simple verb, not associated with any preverb and not preceded by a conjunct particle or a complementizer, and thus standing in absolute initial position in the clause.

- (7) a. *ber-id*                      *cách brith*              *for arele.*  
           bear-PRS.3s.ABS each judgment of other  
           ‘Each bears judgment on the other.’ (Wb 29<sup>b</sup>9; [Newton \(2006: 109\)](#))
- b. *Beogid-ir*              *in spirut in corp.*  
           vivify-3s.ABS the spirit the body  
           ‘The spirit vivifies the body.’ Wb 13<sup>d</sup>7; [Doherty \(2000: 98\)](#)

One salient characteristic of OI is its double system of verbal inflection. The structures with simple verbal forms in absolute initial position display a type of morphology that is referred to as the *absolute* morphology in traditional grammars. A distinct agreement morphology, called *conjunct* morphology, is associated with compound verbs, in configurations where the verbal root is combined with one or several preverbs. The conjunct morphology is also mandatory in the contexts where a simple verb is preceded by a verbal particle, such as negation, the interrogative particle *in*, complementisers and also those where a simple verb is preceded by a semantically empty particle. In other words, conjunct morphology occurs whenever the verbal form is not the initial element in the clause.<sup>3</sup> The verbal morphology found in examples (6) and (7) is thus the conjunct one. The one seen on examples (7a) and (7b) is the absolute. Absolute forms tend to be longer than conjunct ones. For example, corresponding to the absolute form *berid* ‘carries’, we have the conjunct form *beir* found in *ní beir* ‘doesn’t carry’, where the verbal form is preceded by the negative particle, and in *as beir* ‘says’ and *do beir* ‘brings, gives’, where the verbal stem forms a compound with a lexical preverb.

Summarizing: finite verbal forms in OI, at least at certain tenses and moods, display different endings, according to whether they are structurally dependent on or independent from higher peripheral material that c-commands them: conjunct endings appear in the first case, absolute endings in the second.<sup>4</sup> Compound verbs always display conjunct endings.<sup>5</sup>

The initial position in OI clauses has a specific status and at the same time is not restricted as to the formatives it can host. This claim is supported by two additional properties, the distribution of pronominal clitics and the assignment of stress in clusters. In the structures containing a conjunct particle or a preverb, object clitics are found in the second position of the verbal complex, following the formative that comes first, whichever it is, and preceding the verbal form. When the verb is preceded by several elements, the pronoun is systematically affixed to the first one. It can also be encliticized on the dummy

3. On the absolute/conjunct distinction, see [Bergin \(1938\)](#), [Thurneysen \(1946\)](#), [Watkins \(1963\)](#), [Kurylowicz \(1964\)](#), [McCone \(1997\)](#), [García-Castillero \(2020\)](#), among many other philologists.

4. In Modern Irish, only a restricted set of irregular verbs does. [Vendryes \(1908\)](#) observes that the use of absolute inflection in Old Irish was limited by the habit of placing a preverb in front of the verb in certain tenses. The use of this inflection was in fact restricted to the indicative present, future, preterit, and to the subjunctive and imperative moods.

5. In the presentation of the verbal suffixes, we mark absolute suffixes as such explicitly; in contrast, the conjunct suffixes are presented simply as the default forms, with no special marking within the glosses.

particle *no*, whose function here is apparently simply to provide a support for the clitic.<sup>6</sup> In the absence of either a particle or a preverb, it is the finite verb standing in initial position that plays a supporting role. In other words, conjunct particles, preverbs, finite verbs share at least one property when they come first in the linear sequence: they allow weak pronouns to encliticize on them.<sup>7</sup> The fact that the enclitic pronoun always appears in second position in the verbal cluster is known as Vendryes' Restriction. [Strachan \(1909\)](#) provides the following examples.

- (8) a. *beirthi* [berid 'carries']  
 bear.PRS.3s=3s  
 'He carries it.'
- b. *atom chi* [ad-ci 'sees']  
 PV=1s see.PRS.3s  
 'He sees me.'
- c. *condom accae* [ad-ci 'see']  
 C=1s PV-see.PRET.3s  
 'so that he saw me...'
- d. *nom chara* [caraid 'love']  
 PV=1s love.PRS.3s  
 'He loves me.'

The second set of phenomena that support the claim that the initial position has a specific status concerns the placement of stress and the determination of prosodic structure in verbal complexes.<sup>8</sup> Although these complexes can be characterized as phonologically and prosodically unitary structures, with the verbal stem as their core, the place of the main stress suggests that a strong juncture intervenes between the initial element (a conjunct particle or a preverb, potentially followed by an infix clitic) and the remainder of the complex. The relevant cases are recapitulated below (see García-Castillero 2020: 33, 41):

- (9) a. accent on the first syllable of the verbal stem, when it is preceded by a conjunct particle and no preverb is present: *ní beir* 'doesn't bear' ; *ní ric* 'doesn't come' [ni.r(o)-ing];
- b. accent on the first syllable of the verbal stem, when it is preceded by a single preverb: *as-beir* 'says'; *con-ricc* 'comes together, meets' [com.r(o)-ing],

6. Although they are clearly enclitics on the initial element, these pronouns are usually referred to as "infix pronouns" because they intervene between the initial element and the verb.

7. In (8a), the verbal stem in the absolute form is followed by the mark of the 3rd person singular inflexion and by a suffixed 3rd person singular neuter pronominal form; *ci* in (8b) is preceded by the preverb *ad* and the 1st singular enclitic form; in (8c), the pronoun is encliticized on the complementiser *co*; in (8d), it is the dummy particle *no* that provides a support for the clitic.

8. See [Watkins \(1963\)](#), [McCloskey \(1978\)](#), [McCone \(1997\)](#), [Carnie, Harley & Pyatt \(2000\)](#), [Adger \(2006\)](#), [Newton \(2006\)](#) and [García-Castillero \(2020\)](#), for extensive discussion.

- c. accent on the preverb, when the latter precedes the verb and follows a particle or another preverb: *ní for-cmi* ‘doesn’t keep’ from *for-com-óim*; *do-fuisim* ‘creates’ from *\*to-for-ess-semim*.

In cases (9a) and (9b), stress falls either on the second element in the sequence or on the second syllable of a compound verb. In other words, the initial element in the verbal complex is not considered as part of the verbal word for the purpose of stress assignment.<sup>9</sup>

Compound verbs show an interesting behavior with respect to stress. As already observed, they display no variation in their inflectional ending (they systematically bear conjunct inflection). But they have different stems depending on the position they occupy in clausal structure. Preverbs can a priori occupy three different places: they precede stress, bear stress, or follow the stressed syllable. The first preverb of an independent compound verb is unstressed or *pretonic*. The main stress falls on the next element, which can be the first syllable of the verbal root or the second preverb if there is one. The resulting form is then called *deuterotonic*. If a conjunct particle precedes the compound verb, the preverb is included in the stressed part of the verb and stress falls on it. The resulting form is said to be *prototonic*. Note that stress in OI appears to be word-initial generally (as in nouns, for example), so it is the deuterotonic forms that are unexpected. Newton (2007) illustrates this divide with the following examples.<sup>10</sup>

- (10) a. *dobeir in fer in claideb don macc*  
           give.PRS.3s(DEUT) the man the sword to.the boy  
           The man gives the sword to the boy.’  
       b. *ní tabair in fer in claideb don macc*  
           NEG give.PRS.3s(Prot) the man the sword to.the boy  
           ‘The man doesn’t give the sword to the boy.’

## 2.2 What does “first position” mean exactly?

In his celebrated 1963 article, Calvert Watkins observes that when the structure contains no conjunct particle and no complementiser, the first preverb acts as though it occupied the same position as these elements<sup>11</sup> and that in the absence of either a particle, a complementizer or a preverb, it is the finite verb that comes first. Watkins (1963: 30) insists that the distribution of the elements involved is constant and that they can be grouped into classes according to the rank they occupy in the various sequences where they occur. Only

9. McCone (1997: 4) considers the relevant boundary as morphological and defines it as “a kind of barrier or juncture across which certain normal processes do not occur”, morpho-phonological processes such as stress assignment and mutation among them. In root sentences where the verbal complex reduces to a simple verb in initial position, no juncture is present and the verbal form marked for the absolute morphology is assigned stress on the first syllable. This leaves us with a rather complex situation, which McCone (2005: 73-74) summarizes as follows:

...simple verbs retain the same basic form of the stem but typically change the endings from absolute to conjunct upon becoming dependent, whereas compound verbs always have conjunct endings but typically shift from a deuterotonic to a prototonic form of the stem when dependent.

10. In (10), the form *dobeir* “gives” actually consists of the combination of the present 3rd person *berid* of the verb “bring” and of the preverb *do* “to”.

11. Recall that a verbal complex can contain more than one preverb.

three classes are needed - V (finite verbs), P (preverbs), E (enclitic pronominal objects) - because complementisers, conjunct particles and preverbs are mutually exclusive in the periphery and hence all belong to class P.

This conclusion should be handled with care. In Watkins's philological approach, the notion of "position" - the term is used by Watkins - refers to the rank of an element in a sequence, not to a designated site in a grammatical architecture.<sup>12</sup> The major challenge a formal approach has to face is to determine the structural organisation of verbal complexes and, in particular, to provide the notion of "initial position" with a principled status. Several possibilities present themselves. Either the negative particle, complementisers, the first item in a sequence of preverbs *and* simple verbs marked for the absolute inflection all compete for the same initial slot, conceived of as a distinguished site. Or the elements that come first in the linear sequence do not necessarily occupy the same structural site. For example, it must be determined whether a finite verb that is preceded by nothing and is marked for the absolute inflection occupies the same site as a finite verb following a sentential particle and marked for the conjunct inflection. The decision taken on this point immediately affects the characterization of "second position clitics". If the finite verb reaches the highest position in the clause, the position occupied by clitics remains constant when they are affixed to a particle or a complementiser and when they are enclitic to a simple verb. If the finite verb, when it is preceded by nothing, stands lower than conjunct particles and complementizers, clitics, which are always enclitic, are affixed at different levels of structure when a simple verb stands in initial position and when a compound verb or a verb preceded by a conjunct particle is involved.

Formal accounts of the OI double system of verbal inflection, of the proto-tonic/deuteronomic alternation and of the enclitic/infix pronoun divide have been proposed by linguists working within the principles and paramaters and minimalist frameworks, see [Doherty \(2000\)](#), [Carnie, Harley & Pyatt \(2000\)](#), [Adger \(2000, 2006\)](#), [Newton \(2006\)](#), [Newton \(2007\)](#). Although the relevant analyses are not numerous, they strongly differ from one another. There are two reasons for this. First, linguists have made varying choices concerning the realisation site of simple verbs, which, in turn, have different implications for the syntax of clitics. *A priori*, simple finite verbs can either raise to the higher peripheric head when it is not occupied by something else or remain in the lower one; or they can remain in the lower site in all situations. Second, linguists are fully aware that it is difficult, though not impossible, to account for the cliticization facts and for the stress and prosodic facts on a unitary basis. This is why both Adger and Newton argue that a purely syntactic approach to OI verbal complexes cannot be maintained and that additional explanatory principles must be resorted to, namely the ones that govern postsyntactic phenomena, in particular post-syntactic movements. But their analyses differ as to the phenomena that, in their view, should be dealt with in the post-syntactic component.

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12. García-Castillero's (2014) view is different. He distinguishes six slots in the template of the Old Irish verbal complex. Preverbs have access to two slots: the initial one and the one following infixed pronouns. These pronouns, which can precede or follow the finite verb, can stand in slot 2 or slot 6. In other words, the element that comes first doesn't necessarily occupy the initial slot in his model.

While we reject García-Castillero's templatic approach, we embrace his idea that Old Irish grammar makes use of the same grammatical mechanisms as some polysynthetic languages. Our claim is that the mechanism that these very different languages share is multiple head movement, which is central to the grammatical patterns in Algonquian, Slavic and Dene languages ([Branigan 2023](#), [Branigan & Welch 2023](#)); see section 3.1.

The analysis we propose here updates [Carnie, Harley & Pyatt \(2000\)](#), who argue that OI makes use of a long head-movement operation to fill the C-position. We show here that such a movement does take place, but that the motivation for it is to be found in labeling theory and that the mechanisms involved are quite different from what they propose.

### 2.3 The functional sequence in verbal clusters

With these cautionary observations in mind, we can proceed with the syntactic analysis of the patterns that have been isolated in 2.1 and with the identification of the functional sequence that underlies them. Let us first look at the structures in which the finite verbal form is not the initial element.

The clause structure of a language can often be seen more clearly through the examination of embedded indicative clauses, because the basic features of clause structure are often masked by “root transformations” ([Emonds 1980](#), [Emonds 1970](#), [Koster 1975](#), [Den Besten 1983](#)). This is the case in OI, where the presence of overt complementisers provides essential information about what has and what has not occurred in the derivation of the remainder of the clause. In particular, if grammatical subjects in Irish are taken to occupy the specifier of the TP projection, the fact that finite verbs appear to the immediate left of clausal subjects is evidence that verb movement targets a position in the left periphery that is lower than the complementiser. For example, in the embedded purpose clause in (19), the complementiser *arna* precedes the embedded verb *gaba*, which precedes the subject *nech*.

- (11) *as bertar a n-anman arna gaba nech desimrecht*  
 PV say.PRS.PASS their names that.NEG take.SUBJ.3s person example  
*diib.*  
 from.them

‘Their names are mentioned so that no one may take example from them.’  
 (Wb 28<sup>a</sup>20; [Adger \(2006: 617\)](#))

The only conventional cartographic category the realisation site of complementizers can be identified with is Force. The position occupied by the verb at Spell-Out is not involved in any discourse functional roles like Topic or Focus. Rizzi’s (1997) Fin is thus the most plausible candidate to host it.

The same word order is found in the root (and embedded) clauses introduced by the elements referred to as “conjunct particles”. These include the negation markers (*ni*, *na*), the polar interrogative markers (positive *in*<sup>N</sup>, negative *innad*<sup>N</sup>), the relative class markers (*nad*<sup>N/L</sup>, *nach*, *(s)a*<sup>N</sup>, *chech(a)*). The conjunct particles in initial position are endowed with illocutionary force or polarity value. Since [Dedio \(2023\)](#), these particles have generally been considered as complementizers, since they are in complementary distribution with them. If the left periphery of sentences is articulated in several heads, as Rizzi (1997) proposes, and if the relevant heads preserve a minimal semantic content, the highest head in the cartographic schema, namely Force, is an obvious candidate to host these elements. In the corresponding structures, the verb evidently raises to Fin, past the clausal subject. (12a,b) illustrate. (Conjunct verb forms are used in all the clauses which are headed by conjunct particles).



- (12) a. *ni dénat firtu úili*  
 NEG make.PRS.3p miracles all  
 ‘Not all work miracles.’ (Wb 12<sup>b</sup>20; Adger (2006: 610))
- b. *as rubart día friu-som ara celebartis a sollumnu.*  
 PV PRF.say.3s God to-3p that celebrate.SUBJ.3p his feasts  
 ‘God has said to them that they should celebrate his feasts.’  
 (repeated from (6b))

The idea that the finite verb must raise to Fin is supported by the form taken by responses to polar questions, although there are limited textual examples of these. Thurneysen (1946: 479) provides the example in (13), where the response is the direct quotation in the second sentence.

- (13) *in fail nail con desta fil ol Pátraic*  
 Pol exist NEG.other PV seek exist said Patrick  
 ‘“Is there aught else that thou wouldst demand?” “There is”, said Patrick’

OI is a language in which a polar response can be provided by the verb. In Holmberg’s (2015) model of such languages, polar responses are formed through a combination of verb movement to a position in the left periphery with ellipsis of the remainder of TP.<sup>13</sup> If so, then, the OI response verb *fil* in (13) should be situated somewhere within the left periphery, so that it is excluded from the ellipsis of the core (TP) clause. This is consistent with our claim that the verb raises to Fin. The reason why it has to raise will be made explicit in section 3.

The case of pronominal clitics will be carefully reconsidered in section 2.6 below. For the moment, let us note that if we accept the view that the finite verb occupies the lower position in the periphery, there must be an additional complementiser-like position above it to host the initial element supporting the infixed clitic in the structures where the finite verb is not clause-initial and the pronoun is not enclitic on it. The fact that infixed clitics follow the initial element and precede the finite verb (or a preverb if there is one) indicates that there is some space for them in between Force and the FinP projection.

The stress patterns reviewed above point towards the same direction. The Force-Fin analysis makes available a simple characterization of the way stress is projected on the relevant structures: the prosodic juncture stands in between Force and FinP.<sup>14</sup>

We end up with the cartography in (14). Preverbs are included in this schema, but will be dealt with extensively in section 5.

13. That type of ellipsis is clearly attested in contemporary Irish (cf. McCloskey (1996, 2022) and in contemporary Welsh (cf. Rouveret (2012)), even beyond the formation of polar responses. So it is quite plausible that it occurred in earlier Celtic grammars, in particular in OI.

14. Stefan Dedio informs us that there is also a prosodic juncture between preverbal elements and the verb proper in Middle Welsh, but that it is probably not as pronounced as in Old Irish. He suggests that the main reason for this is probably the position of the main stress. “While the main stress in Old Irish lies on the syllable that directly follows a phonologically and morphologically relevant boundary (many phonotactic constraints, phonological processes, and phonological changes are connected with the accent and this morphological boundary), the penultimate position of stress in Middle Welsh (and the ultimate position in Old Welsh) is quite often far removed from this specific morphological boundary, which led to less processes and constraints operating relative to that same position.” Middle Welsh poetry displays some absolute verbal forms, which are absent from Middle Welsh prose. Dedio (2015) endeavours to determine whether they represent an intact absolute/conjunct distinction of the Old Irish type or whether they should be viewed as archaisms.

- (14) [<sub>ForceP</sub> [<sub>Force</sub> conjunct particle/complementiser/preverb] - Cl - [<sub>FinP</sub> [<sub>Fin</sub> preverb [V-Fin(CONJ)]] ... ]

Let us now look at the clauses in which the finite verbal form is not preceded by anything. They display none of the properties associated with complex verbal clusters in initial position. As already mentioned, the simple verbs in these structures are endowed with absolute inflection. They systematically bear stress on the first syllable, which is the unmarked case in OI. Object clitics are enclitic on the verb, they never precede it. The following examples are repeated from (7a) and (8a).

- (15) a. *berid cách brith for arele*  
 bear.PRS.3s(ABS) each judgment of other  
 ‘Each bears judgment on the other.’  
 b. *beirthe* [*berid* ‘carry’ ]  
 bear.PRS.3s(ABS)=3s  
 ‘He carries it.’

As already proposed by Doherty (2000), these properties suggest that the clauses with a simple verb in initial position have a structure different from the ones involved in sentences beginning with a full-blown verbal cluster. The latter must be ForceP projections, the former behave as if they reduced to FinP projections. This can mean either that Force is not projected at all or that it is invisible or inactive. The cartographic representation of the relevant structures is as follows.

- (16) (Force) [<sub>FinP</sub> [<sub>Fin</sub> V-tense-agr] [<sub>TP</sub> DP<sub>subject</sub> ...

We have thus justified two basic structures for OI finite structures, namely (14) and (16).<sup>15</sup> We take Force to be the equivalent of Chomsky’s (2015) and Carnie, Harley & Pyatt’s (2000) C category. In our approach, the landing site of OI finite verbs is the lower periphrastic head Fin in all structures (abstracting away from the tmesis construction, see next subsection).

## 2.4 The relevance of labeling

In labeling theory, features, not categories, provide projections and complex syntactic objects with labels. Let us provisionally assume that conjunct particles and complementisers, which are externally merged at the Force level, bear a periphrastic feature, which can be a discourse feature, an illocutionary feature and/or a polarity feature, and that this feature provides the whole clause with a label.

The structures beginning with a simple verb also deserve careful attention in this respect. Force is the initial home of tense and agreement features, which, in many languages, are transferred through Feature Inheritance to a lower host. This host has been identified with Tense (T) in English and French. In OI, however, Feature Inheritance copies  $\phi$ -features from Force to Fin, the position to which the finite verb raises from

15. In section 4, *wh*-questions will be shown to require an additional one.

A crucial observation is that substituting a conjunct form (*beir*) for the absolute one (*berid*) in (15a) and (15b) would give rise to ungrammatical sentences. A natural explanation for this restriction is that absolute morphology plays a direct role in the labeling of the clause, just as rich inflection in pro-drop Romance languages does, according to Chomsky (2013). Our hypothesis, which will be refined in section 2.5, is that the absolute inflection bears the whole gamut of  $\varphi$ -features, that, taken together, these features constitute an interpretable unit and that this unit, which occupies the highest head position, functions as the label of the whole clause. The ungrammaticality in the same context of verbal forms endowed with conjunct morphology indicates that this morphology is not, in and of itself, able to provide a label for the domain it heads. Raising to Fin is not sufficient in this case.

(17) a. *ad-cruth cain cichiter*  
PV-form fair see.FUT.PASS.3s  
'Fair form will be seen.'  
b. *ad-cichiter cruth cain* (normal form)

Tmesis constructions are not the only exception to the verb-initial requirement in OI. The syntactic pattern known as Bergin's construction, which is attested exclusively in poetry and in high register (law) texts, also constitutes a departure from it.<sup>16</sup> In Doherty's view, this construction reduces to a verb-second construction.

- (18) a. *Lugaid Luath loisc trebthu trin tuath*  
 Lugaid Swift burn.PRET.3s dwellings strong people  
 ‘Lugaid the Swift burned the dwellings of strong people.’  
 (Doherty (2000: 99))

11

- b. *maicne nailb 'airmi*  
 sons.ACC Alb.GEN reckon.PRS.2s

‘You reckon the sons of Alb.’

(Bergin (1938: 201))

The initial phrase is a subject in (18a), an object in (18b); the verb is in medial position in (18a), in final position in (18b). Doherty’s general conclusion is that early Irish was a residual verb-second language and that both *tmesis* and Bergin’s construction should be analysed as verb-second residues.

A question that far exceeds the limits of this article is which derivational path allows Bergin’s construction and more generally verb-second constructions to be appropriately labeled.<sup>17</sup> Let us provisionally assume that the presence of a phrase in initial position in V2 constructions is a key dimension in their labeling. As the glosses indicate, the verbal forms in Bergin’s construction are marked for the conjunct morphology. This is the case, for example, in construction (19), headed by an interrogative pronoun, which can be analysed as a residual verb-second structure.

- (19) *cia beir búar o thig Temrach?*  
 who take.PRS.3s cattle from house Tara

‘Who brings cattle from the house of Tara?’

(LL 1566; Newton (2006: 166))

(20) shows that, as far as verbal morphology is concerned, this construction aligns with the structures beginning with a verbal-complex. This behavior conforms to the characterization of the conjunct morphology given above.

In this section, the linear organisation of  $X^0$  heads in OI verbal complexes and in simple verb constructions has been considered. It was argued that the two should be distinguished. The complex verbal clusters define ForceP projections; the sentences that begin with simple verbs are undistinguishable from FinP projections.

## 2.5 The absolute/conjunct asymmetry and the phasal/non-phasal status of the Fin head

As already discussed, the choice between absolute and conjunct inflections depends entirely on the morphosyntactic environment. There are two conditions which must be satisfied for absolute inflection to be used: the verb must not be preceded by a conjunct particle or a complementiser and the verb must not follow a preverb.

- (20) *Beogidir in spirut in corp.*  
 vivify.PRS.3s(ABS) the spirit the body

The spirit vivifies the body.’

(Doherty 2000: 5))

These two constraints cannot readily be collapsed without careful consideration of what deeper grammatical principles they reflect, so let us consider them one at a time.

We maintain the hypothesis that the OI finite verb, whether absolute or conjunct, always raises to Fin. The first generalization to be derived is thus the following:

17. For preliminary suggestions concerning the labeling of verb-second constructions, see Branigan (2020), Rouveret (2023).

- (21) Fin expresses absolute inflection as long as Fin is the leftmost element within the clause.

Consider now what it takes for Fin to be leftmost. Minimally, this means that there must be no Force complementiser present. If we continue to assume that it is the Force head which introduces the unvalued  $\varphi$ -features that are later transferred to Fin, then the absence of Force can be attributed to the complementiser deletion operation (Chomsky 2015), which applies at the phase edge only if C/Force transfers all its formal features away.

Now, in our approach, the deletion/retention of Force is entirely dependent on the necessity of a label for the interpretation of every phrasal category. When Force is a visible complementiser or hosts a negative or interrogative verbal particle, the feature it bears will automatically serve as the label for ForceP.<sup>18</sup>

But the labeling status of the root clauses with a silent Force head is less obvious. Suppose that Force has transferred (almost) all its featural content to Fin, then what is left in the Force position is presumably insufficient to serve as a label. In that case, Force deletion can be an operation forced to prevent a labeling failure. When the phase head deletes in this way, the head of its original complement, to which it will have transferred some formal features, acquires a *derived* phase head status as a result Chomsky (2015). The implication for the OI structures is that deletion of the original Force phase head will cause Fin to serve as a derived phase head. And this then allows us to refine the generalization that we need to capture. The revised version is the following:<sup>19</sup>

- (22) Fin expresses absolute inflection if and only if Fin is a phase head.

(23) allows us to clarify the way phase structure affects the computation of labels. In conjunct verbal forms, the features of Fin are contained in the portion of the clausal phase within which tense, aspect and the verb are made ready for interface interpretation. As for the agreement features, which are also present on Fin, they are valued and deleted within this domain, if one reasons within Chomsky's (2008) probe-goal-Agree approach to syntactic dependencies. As a consequence, the features of Fin that provide a label cannot be the agreement features and must be the interpretable features found on the Fin head, such as tense and mood features. The case of absolute forms is different: they require that Fin be a phase head. This means that the agreement features in Fin cannot be immediately processed and deleted (within the FinP phase), but must persist until the next phase is complete. They are therefore available for labeling purposes in absolute forms. But this can only happen if the  $\varphi$ -features are rich enough, just as a single  $\varphi$ -feature set provides a label for the clause in Italian, but not in English (Chomsky 2015).<sup>20</sup> The "richness" of  $\varphi$ -features cannot be determined simply by looking at the affixes provided in the Fin position. It must also take the verb stem into account, given the complexity of

18. The same result obtains when the clause-initial position is occupied by a preverb which, as shown in section 5, raises to Force.

19. It is not our concern here how exactly the connection between the phase head status of a category and its morphological realisation is made concrete. This can certainly be modeled with DM resources, but we suspect that other models can accomplish this goal as well.

20. The inflectional characteristics of OI verbs are far more involved than in English or Italian. Among other things, OI inflection distinguishes deponent verbs from simple active verbs (Thurneysen 1946). And future and preterite suffixes *for some verbs* appears together with a reduplicant prefix, which affects the suffix form.

the OI verb conjugations. So richness of inflection in Fin must be calculated on the basis of the pairing of a verb stem with inflectional features. In the approach adopted here, the combination of an inflectional affix with a verbal stem is mediated by [stem] features.<sup>21</sup> Since Fin, at the phase level, contains the [stem] feature that identifies the verb which will eventually combine with the affix, all the information necessary to evaluate the richness of the inflection is available at the point of labeling. In (20), for example, the affix in Fin will contain the 3SG  $\varphi$ -features and the stem feature [stem(beogid-)]. Together, these features allow a rich label to project.

It still must be explained why preverbs and absolute inflection are mutually exclusive. Why is it that the raising of a preverb to Fin and its adjunction to an absolutely marked verbal form doesn't give rise to a well-formed representation? Remember that absolute verbal forms qualify as phase heads in Fin. And it will be shown in section 5, prototonic preverbs are attracted to Fin before the verb. The generalization stating the incompatibility under discussion can thus be encapsulated in the following statement.

- (23) A Fin head which is a phase head cannot attract multiple heads.<sup>22</sup>

The question now becomes why multiple head movement in OI should be constrained in this way. Affixes in Fin bear the number of [u-stem] features necessary to attract all the preverbs present in the clause and the verb itself. The [stem] features in Fin, once valued, drive the movement of  $X^{\circ}$  heads.

When preverbs are included in the structure, however, the affix must contain more than one [u-stem] feature and valuation operations will then be faced with several distinct [stem] features. In this case, there is no obvious way to determine which [stem] feature matters for determining that  $\varphi$ -features are suitable for labeling. It follows that  $\varphi$ -features on a phase head like this should not be able to project, which means in turn that absolute verbs must not be associated with preverbs.

To summarize, absolute verbal inflections are employed when Fin is the phase head. This typically means that nothing else can be present in the higher left periphery. The exclusion of preverbs from absolute verbs follows from the way the richness of  $\varphi$ -features for labeling is computed in the derivations where Fin is a phase head. This leaves open the status of deuterotonic structures, where the initial preverb does not join tightly to the rest of the verbal complex. These structures arise when a preverb raises and adjoins to Force, as discussed in section 3.2.

## 2.6 Infixes and suffixed pronominal clitics

The analysis developed here for the absolute/conjunct (prototonic and deuterotonic) forms is based on the premise that Force, when it is filled, qualifies as a phase head, but that it can give up its phase head status to Fin through Force deletion. The distribution of object clitics in OI is immediately explicable within phasal structures of this type. This

21. More will be said about [stem] features in section 5.1. For the moment, it suffices to know that they are directly relevant to the combinations of  $X^{\circ}$  heads into  $X^{\circ}$  clusters. Although they differ from formal syntactic features, which deal with the dependencies that underlie the order of clausal constituents, they share some properties with them: they come in two varieties (interpretable vs. uninterpretable); they must be valued when uninterpretable.

22. Multiple Head Movement is provided with a theoretical footing in section 3.1.

could open the way to a novel approach to the “Wackernagel’s effects” found in many languages, which include both object personal pronouns and relative markers in a subset of structures.

The short and simple description of clitic placement in OI is as follows. Clitic pronouns are always moved to the immediate right of the phase head of the domain they belong to.<sup>23</sup> As seen earlier, object clitics are found right after overt complementisers, which always serve as phase heads.

- (24) a. *nís nágathar.*  
NEG=3p PV-fear.PRS.3s  
‘He does not fear them.’ (Thurneysen (1946: 255))
- b. *ním charatsa.*  
NEG=1s love.PRS.3p.CONJ-EMPH.PRT.1s  
‘They do not love me.’ WRONG REFERENCE??
- c. *nís (n)imdich.*  
NEG=3p PV-protect.PRS.3s  
‘He does not protect them.’ (McCone (1997: 11))

Object clitics also appear to the immediate right of the initial preverb, which is adjoined to the silent phase head in deuterotonic structures.

- (25) *dus ngni*  
PV=3s.F make.PRS.3s  
‘He makes it.’ (Ml 29 3; Newton (2006: 31))

They are also found immediately after absolute verb forms (with certain caveats), as in (26) (See also examples (7)).

- (26) a. *bertaighthius*  
shake.PRS.3s(ABS)=3p  
‘He shakes them.’ (McCone (1997: 23))
- b. *comallaidi*  
fulfill.PRS.3s(ABS)=Masc.3s  
‘He fulfills it.’ (Ml 94<sup>b</sup>1; Adger (2006: 618))

The derivational mechanism for clitic placement can hardly be determined on the basis of the OI data alone. But comparison with the distributional behaviour of object clitics in other languages could open the way to a reasonably general treatment. It must be emphasized, however, that the situation instantiated by clitic pronouns in OI is quite

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23. Noonan (2011) also pursues an analysis of the Old Irish clitics within phase theory, but based on different premises. For Noonan, the presence of an object clitic introduces a new phase boundary between C and T, which blocks movement of the verb in T to the C position. This approach does not extend naturally to cover the deuterotonic structures considered in section 5.



unexpected if one reasons along the lines of the familiar accounts of the enclisis/proclisis divide, which have been proposed for example for pronominal clitic placement in Romance. Since at least [Kayne \(1991\)](#) and [Rivero \(1994\)](#), the enclisis of pronominal clitics (in Romance imperative and infinitival clauses and in European Portuguese root clauses) has been taken to be the consequence of the movement of the verb to a position higher than the one hosting the clitic. The postverbal position of the clitic (or of the clitic cluster) can be derived from the raising of the verb, provided that clitics are first merged in an inflectional position higher than that of the verb, as Kayne (1991) suggests. However, several authors have shown that this geometric approach to cliticisation meets with serious difficulties (on European Portuguese root clauses, see [Rouveret \(1992\)](#), [Nash & Rouveret \(2002\)](#)). Kayne's approach predicts that the verbal forms supporting an enclitic are higher than the ones preceded by a proclitic. Adverbial distributions and the positions of subjects definitely show that the verbs occupy the same position in both cases. Contrary to the predictions of Kayne's account, the relation between enclitics and the verb turns out to be much tighter than the relation between proclitics and the verb. In European Portuguese, proclitics can be put in common factor across the two members of coordination structures, whereas enclitics must be repeated on each verbal form; in the modern period, an adverb was allowed to intervene between a proclitic and the verb (cf. [Rouveret \(1992\)](#)). It is interesting to observe that what has been proposed in the preceding sections concerning the syntax of verbs in OI does not fit Kayne's characterization, but displays one of the defining features of European Portuguese: the finite verb occupies the same site when it supports an enclitic and when it supports a proclitic. In OI, the finite verb, whether it is marked for the absolute or the conjunct morphology, occupies the Fin position. What is relevant to the infix/suffix contrast is not the height of the verb, but its being c-commanded by/dependent on higher material or not. We also note that the syntactic space potentially available to clitics is more extended in conjunct verbal complexes than in constructions with an absolute verb form. Since the former are ForceP projections and the latter FinP projections, the clitic could adjoin either to Force or to Fin in the former, but can only adjoin to Fin in the latter. In conjunct constructions, only one choice is available: the clitic is encliticised to the content of Force and that in absolute constructions, it can only be encliticised to the verb in Fin. The two OI constructions share at least one property: the clitic pronoun is affixed as an enclitic to the category that qualifies as a phase head.

Two questions arise at this point. First, it must be explained why pronominal clitics are never the initial elements in the sequence. This follows if the grammar incorporates the following constraint (Ouhalla's (2008) formulation).

- (27) A pronominal clitic cannot be the first head constituent in the minimal domain (CP, DP, or PP) that includes it.

(27) encompasses the so-called Tobler-Mussafia's Law which, in Romance, prevents pronominal clitics from standing at the absolute beginning of clauses. In Ouhalla's approach, the first head in a domain is the one that is not c-commanded by any other head in this domain. A pronoun encliticized on either Force or Fin is not the first element. One could ask why such a constraint holds at all. Labeling Theory could provide the answer. For some reason, a clitic pronoun cannot function as the label of the domain



that minimally contains it. It is clear that the value of the  $\varphi$ -features borne by an object clitic will not necessarily coincide with the value of the  $\varphi$ -features of the head it affixes to. If feature-sharing presupposes sharing of values, not sharing of attributes, the Labeling Algorithm cannot be resorted to to redeem clitic-initial structures.

Second, it must be understood why clitics entertain a special relation with phase heads. Only a tentative answer can be given at this point. By definition, phase heads are the categories that initially host the formal features that determine the nature and the form of dependencies within syntactic domains. Uninterpretable  $\varphi$ -features are among the specifications that originate in Force. In ForceP conjunct structures, these features are indeed transmitted to Fin, but since Force is not deleted, they remain accessible on Force until the phase is completed. In FinP absolute structures, Force is deleted after feature transmission and  $\varphi$ -features only show up on the derived phase head Fin. The reason why object clitics are sensitive to the presence of  $\varphi$ -features on Force or on Fin is that clitics can be thought of as elements that are *unselectively* attracted by heads endowed with such features.<sup>24</sup> The phrase “unselective attraction” is intended to capture the idea that, when they move, clitics don’t satisfy a selectional requirement of the category they attach to (selectional requirements have been evaluated before cliticisation), but take advantage of the presence of a head with  $\varphi$ -features and are “attracted” by this head, although the  $\varphi$ -values do not coincide.

For reasons that remain obscure, pronoun enclisis to the absolute verb is limited to particular verb classes, and further constrained by the agreement inflection and the choice of the enclitic object. For most verbs, enclisis is limited to 3rd person objects and to verbs with 3rd person subjects or verbs with 1st person plural subjects in the future tense [Thurneysen \(1946: 271\)](#). What is more, when the subject is not a 3rd person singular, a masculine and neuter clitic is expressed allomorphically as *it*. ] When these constraints are not observed, the absolute form gives way to a conjunct verb preceded by the expletive preverb *no*, as in (28).

- (28) a. *nos comalnammar*  
PV=3p fulfill.PRS.1p  
‘We fulfill them.’ ([McCone \(1997: 12\)](#))
- b. *nom charat*  
PV=1s love.PRS.3p  
‘They love me.’ ([Thurneysen \(1946: 348\)](#))
- c. *nom isligur*  
PV=1s abase.PRS.1s  
‘I abase myself.’ (Wb 17<sup>b</sup>22; [Newton \(2006: 23\)](#))

In fact, as McCone notes, “even where suffixing of pronouns is common or normal, infixing [*sic*] with *no* is an optional possibility”:

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24. On cliticization as unselective attraction, see [Nash & Rouveret \(2002\)](#), [Rouveret \(2020\)](#).

- (29) *nos mbertaigedar*  
 PV=3s shake.3s  
 ‘He shakes it.’

## 2.7 Relative and interrogative structures

### 2.7.1 Relative markers

In relative clauses and in cleft sentences, both conjunct and absolute verbs appear with a *relative marker*.<sup>25</sup> The relative markers also show up in *wh*-questions [García-Castillero \(2020\)](#), which are briefly considered in section 2.7.2 In relative clauses and clefts, the relative marker appear to function as a relative pronoun operator, as in (30b), from [García-Castillero \(2020: 107\)](#).

- (30) a. *pridchimmi* [pridch-immi]  
 preach.PRS.1p  
 ‘We preach.’ (Wb 8b12)
- b. *is crist pridchimme*  
 be.1s Christ preach.1p.REL  
 ‘It is Christ that we preach.’ (Wb 14d40)

In the cleft structure in (30b), an enclitic relative suffix is adjoined to the absolute verbal form (30a).

The attachment of relative markers to verbal forms mimics that of enclitic object pronouns. They can only attach to 3rd person verbs, and in early manuscripts, to 1st person plural verbs. Morphologically, the relative marker is expressed together with the absolute finite inflection: it is right-adjoined to Fin, the phase head.

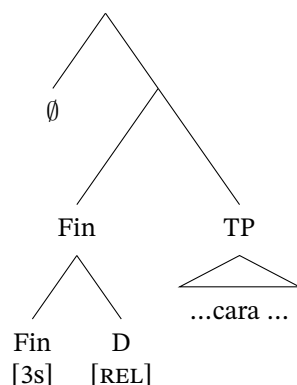
- (31) a. *caraid*  
 love.PRS.3s  
 ‘he loves’
- b. *caras*  
 love.PRS.3s.REL  
 ‘whom he loves’

The structure for the relative form in (31b), before any movement takes place, is as in (32).

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25. Some traditional descriptions use the term *relative marker* to refer only to the affix attached to absolute verbs. We find it convenient to employ the same term for the leniting component which appears in conjunct forms, because the syntactic effect seems to be the same in both cases.

(32)



Adjunction of the relative marker to Fin here constitutes an A-bar movement operation, which creates an operator-variable chain, just as *wh*-movement does.

With simple verbs which are inflected with other agreement features, in the absence of preverbs, an expletive preverb *no* must be adjoined to Force, so that the relative marker may be expressed as an enclitic above Fin. In this case, the phonetic form of the relative marker is different—it is realised through lenition of the following verb or preverb, as in (33).

- (33) a. *caraim*  
love.PRS.1s  
'I love'
- b. *no-∅<sup>L</sup> charim*  
PV-REL love.PRS.1s  
'whom I love'

These forms can readily be accommodated into our model. As always, the verb raises to Fin. The relative marker must also raise to the left periphery, like other A-bar operators. Since it is enclitic, it cannot occupy a specifier position and it can only be attracted to the phase head. When Fin is the (derived) phase head, the relative marker attaches to Fin, as in (30b) and (31b). But when Force deletion does not take place, as in (33b), Force is the phase head to which the relative marker is attracted. In that case, the clitic operator takes the form of the leniting item, which can be seen only in the effect it has on the verbal word in Fin.

### 2.7.2 Two strategies for *wh*-questions

We now turn to another set of configurations in which the phase head status of categories in the left periphery is critical to the understanding of surface forms. Bergin (1938) observes that there are two strategies to form constituent questions in OI (see also Doherty 2000, Newton 2006). The first one consists in building the interrogative sentence as a cleft construction. The cleft sentence pattern displays a stressed interrogative followed

by a relative clause and by a verb in the relative form. Doherty provides the following examples.

- (34) a. *cía ranna=s dúib?*  
 who divide.3s=REL for.you  
 ‘Who divides for you?’ (LL 113<sup>b</sup>12; Doherty (2000))
- b. *cid as dénti?*  
 what be.PRS.3s.REL to.be.done  
 ‘What is it which is to be done?’ (Wb 12<sup>d</sup>41; Doherty (2000: 101))

Relatively-marked verbal forms can be considered as akin to absolute ones. They behave as independent forms, as if the relative boundary sheltered them from the interrogative element. The relative clause constitutes a phase of its own, excluding the *wh*-expression.

The second strategy is exemplified by non-cleft *wh*-questions in which the *wh*-phrase is an unstressed interrogative pronoun that behaves like a conjunct particle. In these structures, the finite verb doesn't take the relative form, but the conjunct or pretonic one, which leads Bergin (1938: 205) to claim that the verb and the pronoun that immediately precedes it are included in the same clause. In more contemporary terms, this means that a single phase is involved, which includes both the finite verb and the *wh*-clitic. The latter appears at the left edge of the interrogative clause, indicating that *wh*-movement has taken place.

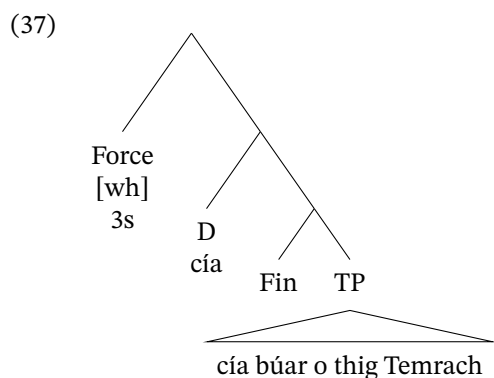
- (35) a. *cía beir búar o thig Temrach*  
 who take.3s.CNJ cattle from house Tara.GEN.s  
 ‘who takes cattle from the house of Tara?’ (LL 1566; Bergin (1938: 205))
- b. *cía acca?*  
 who see.PST.2s  
 ‘Who did you see?’ (LU 135; Bergin (1938: 206))
- c. *co acci in-slúag* [co-ac-ci]  
 how PV-see-2s the-host.ACC  
 ‘How do you see the (fate of the) army?’ (GOI: 290; García-Castillero (2020: 197))

Non-cleft questions in Ol are only found when the interrogative element is an unstressed, monomorphemic pronoun, they are excluded with *wh*-phrases combining a *wh*-element and a noun. This limitation is reminiscent of Chomsky's (2008) discussion of the labeling properties of the structures in (36).

- (36) a. I read what you wrote.  
 b. \*I read which book you wrote.

Chomsky infers from this contrast that simple *wh*-heads and *wh*-phrases do not make equal contributions to clausal labeling.<sup>26</sup> Since *what* in (36a) is a bare interrogative D, it serves as the label for the complement of *read*. But in (36b), the *wh*-phrase is actually phrasal, and cannot itself project. The ungrammaticality of (36b) then stands as a violation of the labeling requirement (the exocentric pairing of *which book* and *you wrote* remains unlabeled).

This reasoning can be adapted to the OI constraints. The unstressed interrogative pronouns in (35) are  $X^0$  categories which can serve as labels. Thus, in (35a), the structure after *wh*-movement has taken place is (37).



Here, the D which has undergone *wh*-movement is not actually *cía* yet, of course, since the phonetic substance of a pronoun is presumably established post-syntactically, as part of the process of externalisation. Instead, *cía* represents a cluster of ph-features, and nominative Case. The origin of the *wh*-feature, however, is the Force head, as in other questions. Force transmits its *wh*-feature to the displaced pronoun, which is the inevitable head of its complement, given Minimal Search constraints. The addition of *wh*-features to the inherent  $\varphi$ -features of the pronoun will ensure that this is pronounced as an interrogative pronoun. On the semantic side, the same *wh*-feature will provide the usual interpretation of the upper copy of the *wh*-phrase as an operator binding a variable.

As in the English case, the transfer of a *wh*-feature away from Force will lead to Force deletion. In this case, however, Fin does not become the new phase head in these structures, because the interrogative pronoun is a higher head to which Force has transferred features. As a result, the verb raised to Fin will not take the absolute form used for derived phase heads. The verb can only take the default conjunct form in questions with unstressed *wh*-phrases.

The structure in (37) raises the question of how the unvalued agreement features which are introduced in the Force head are transferred to Fin in this type of *wh*-question. The agreement features are not realised on the interrogative pronoun, so they are not transferred together with the *wh*-feature. Our solution here is taken from the *multiple feature inheritance* model of Branigan (2020), in which a series of Feature Inheritance operations is possible, transferring features from a phase head to successively more remote target heads, with the final one transferring uninterpretable features such as the

26. What Chomsky actually proposed at the time is that the former project, the latter do not.

$\varphi$ -features expressed as agreement. In (36), Force will transfer the *wh*-feature to *cía* and then the agreement and tense features to Fin.

When the conjunct structure is forced by the use of an unstressed interrogative pronoun, the object pronoun, when there is one, appears to the right of the interrogative pronoun, as shown in (38).

- (38) a. *Cich =ib foruireth?*  
           what=2s PV-do.3s.PFV  
           ‘What has been done to you?’ (TBF 334; [García-Castillero \(2020: 196\)](#))
- b. *cich =e goin?*  
       who=3p kill.PRS.3s  
       ‘Who kills them?’ (TBC-1<sup>2</sup> 1215; [García-Castillero \(2020: 196\)](#))
- c. *ar-fess-id                      cia =b de-r-géni*  
       PV-know-2p.SUBJ what=2p PV-PFV-make.3s  
       ‘you may know who has made you’  
           (Ir.Gosp.Thom. 6; [García-Castillero \(2020: 196\)](#))

This follows from the presence of agreement features in the conjunct verb in Fin. Fin can unselectively attract the object clitic in this case, as in the other conjunct forms where clitics are found. This ensures that the object clitic will be located between the *wh*-word and Fin.

As the model predicts, stress assignment and the associated phonological processes of lenition, eclipsis, syncope, *etc.* will apply in these particular *wh*-question structures in the same way as with overt complementisers: stress will fall on the initial syllable of the verb in Fin.

### 3 Old Irish preverbs and Multiple Head Movement

Up to now, we have carefully avoided fully analysing the preverbs and their syntactic behavior in clusters. It is fair to say that they raise additional difficulties beyond those that have been met in the various constructions studied in the preceding sections. Their salient characteristic is that, contrary to verbal forms, they can occupy different sites in the clausal architecture, and, contrary to conjunct particles and complementizers, their position at Spell-Out is the result of an Internal Merge operation, not of an External Merge one. Here we show that most of the principles and mechanisms that are necessary to account for these features are already at our disposal, but that, in order to achieve a fully satisfactory account, an additional notion has to be introduced, that of Multiple Head Movement.

#### 3.1 Multiple Head Movement

A salient characteristic of OI verbal complexes is that they are made up of several functional and lexical units which originate in the lexical or the inflectional domains of the clause, unlike the conjunct particles and complementizers, which are externally merged

in their Spell-Out position. Neither preverbs, nor the verbal root, nor inflectional suffixes, nor object clitics originate in the clausal periphery. We account for the existence of derived verbal complexes with a mechanism that is able to create a complex unit *via* the raising of several heads to the same category. This word-formation process is what Collins (2002) calls Multiple Head Movement (MHM). Here we employ the model of MHM developed and documented in Branigan (2023). MHM displaces several heads into a single word-sized landing position by joining them with an attracting higher head. This type of movement is not immediately compatible with the standardly accepted idea that the main grammatical constraint controlling head-movement is Travis’s (1984) Head Movement Constraint (HMC), but Branigan (2023) examines structures in several languages in which the standard HMC appears to be inactive and demonstrates that the corresponding grammatical systems are precisely those where multiple head movement is possible.

These results are encapsulated in a general model of head-movement which we briefly summarize here. Head movement involves two components, a [u-stem] feature valuation operation which takes place within the core syntax at the phase level and a substitution operation (referred to as “head-movement” in Principles and Parameters frameworks) which operates as part of the externalization process, forming complex words from atomic roots and affixes. Valued [stem] features are visible and interpreted only in the PF component. They differ in this respect from semantically significant formal syntactic features ( $\Phi$  features, discourse features, *etc.*) But they also share properties with them: they undergo feature valuation and Feature Inheritance and they are initially interpretable on some categories, uninterpretable and unvalued on others.<sup>27</sup> Interpretable stem features are lexical properties of roots and stems, including OI preverbs. Unvalued [u-stem] features are present on categories which trigger head-movement. In this model, regular head-movement is causal reflex of the [u-stem] valuation operation which takes a feature from a root and provides it to the c-commanding affix. Constituents with matching [stem] feature values are interpreted as words for morphophonological processes.

This theory of head movement and of word formation relies on constraint (39), which, although not made explicit, was relevant to our treatment of the absolute/conjunct divide in section 2.5.

(39) *Stem Feature Constraint*

Terminal items which contain the same [stem] feature must be realised within a single  $X^0$  category.

Constraint (39) simply characterises the conditions which allow valued [stem] features to be successfully interpreted for the morphophonology. This is also what drives the head-movement operation which takes place as part of externalization. This head-movement takes place too late to affect the core syntactic structure. But this does not

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27. The system that is proposed here makes use of both [stem] features and formal syntactic features. It is plausible to assume that they interact in some ways and some discussion of how they do so would be welcome. One option worth considering is that in the languages such as OI which resort to MHM, Fin’s feature matrix carries multiple [u-stem] features but one of them has a specific status: it is inherently coupled with a formal feature. In verbal complexes, the [u-stem] which is coupled with a formal feature is the one which triggers the raising of the finite verb and the formal feature it is coupled with is [ $\Phi$ ]. One can think of the valuing process in these structures as similar to the way Case features in English are valued in Chomsky’s (2000) approach: uninterpretable  $\Phi$ -features on T are valued via subject raising and the uninterpretable nominative Case features on T and on the raised nominal subject are valued as a bonus. Similarly, it could be argued that the valuation of Fin’s formal  $\Phi$  features in OI is parasitic on the valuation of one of its [u-stem] features.



preclude the possibility of earlier syntactic operations which displace simplex  $X^0$  constituents within the syntax. Such operations cannot be driven by the same constraint (39), since they would actually *bleed* the environment for [u-stem] feature valuation, as discussed below (section 52a).

Constraint (39) is accompanied by the parametric choice (40).

- (40) *Valuation of [u-stem]*  
Valuation of [u-stem] in F follows/precedes LA.

As proposed by Müller (2009) and Lahne (2008), languages may differ in the timing of the valuation operation at the phase level. In some languages, [u-stem] valuation takes place after labeling; in others, it precedes it. When labeling precedes [u-stem] valuation, the projected label is the only available goal for the [u-stem] bearing probe. But when valuation precedes labeling, multiple head movement is possible, because the probe may look further into the complement of the head to find one or more matches.

As shown in Branigan (2023), the setting of parameter (40) can differ for different probe categories, just as other parameters may apply differently in different categorial domains (Roberts 2020). This allows multiple head movement to take place to some affixal heads in Old Irish, but not to others.

Summarizing, head-movement is a grammatical process which belongs to the system of externalization, but which is driven by prior, phase-level valuation operations, operating on [stem] features. While the head-movement operation may operate identically in all languages, the [u-stem] valuation operation produces different results, based on the values of the pertinent valuation timing parameters, which controls whether [u-stem] features are valued before or after labeling takes place. If labeling precedes [u-stem] valuation, then the result is movement of a single head, (indirectly) subject to the HMC. But if labeling follows [u-stem] valuation, then the result may be movement of several heads to a single probe position, resulting in what looks superficially like a “tucking-in” pattern of head-movement.

### 3.1.1 The morphosyntax of preverbs

Preverbs, the preposition-like lexical items which appear before the verb and form with it what is usually identified as a “compound”, are a central concern in any treatment of OI syntax. An understanding of their core properties must be in place before the derivational procedure can be made explicit. Preverbs appear to the left of the verb, either singly or in combination with other preverbs. They are fundamentally implicated in the verbal lexical semantics, in a way reminiscent of Germanic verb-particle structures: the combination of a preverb with a verb often produces unpredictable, non-compositional meanings for the verb phrase as a whole.

Some examples of preverbs and how they combine are listed in (41), taken from García-Castillero (2020: 38). The related prepositions are shown with a superscript L or N that indicates whether they lenite or eclipse a following noun. This feature is important in understanding the surface form of “compound” verbs and verbal nouns.



- (43) *Ath (mór)cathu      fri crícha    comnámant      -cuirethar*  
 PV (great).battalions to borders neighbour.GEN.p send.3s

‘He dispatches (great) battalions to the borders of hostile neighbours.’

AM 15; (García-Castillero 2020: 84)

The preverb here is *ath*, which semantically combines with the verb *ceird* to produce the lexical meaning “dispatch” for the verbal predicate as a whole. The preverb is separated from its associated verb by one or more full phrases, quite a long one in (43).

Finally, as the existing literature recognises, a compounding analysis raises challenging questions concerning the *deuterotonic* stress pattern which arises when the preverb is the initial  $X^0$  component in the verbal cluster, and does not display the morphophonological characteristics otherwise found with morphemes contained within the lexical core, such as lenition, eclipsis, and other segmental phonological processes.

### 3.1.2 Preverbs in verbo-nominal constructions

A reasonable starting point for the analysis of preverb-verb pairs is the verbo-nominal construction, where the word order and other complexities of clause structure can be kept largely to the side. The reason is that verbal nouns, contrary to finite forms, do not raise to the clausal periphery<sup>28</sup> and are not found in the inflectional domain either, except perhaps in a restricted set of structures.

Verbal nouns are nominal forms of the verbal paradigms, which Celtic languages use to express nonfiniteness in contexts where English typically resorts to infinitival or participial or gerundive forms.<sup>29</sup>

Verbal nouns are inflected with morphological case and mark their nominal complement with the genitive. They are formed via the adjunction of a nominalizing affix to the verbal stem, as shown in (44). Several inflectional classes can be distinguished, which are in part determined by the final element of the root. (44) also shows that a preverb can precede the verbal root. The meaning of the derived nominal retains the non-compositional meaning of the verbal counterpart based on the same root.

- (44) *ni    epur    etar-scar-ad      fri suidiu*  
 NEG say    PV-separate-VN.ACC

‘I say not to you to separate from these.’

Wb 9<sup>b</sup>19; Thurneysen (1946: 445)

Now consider (45), where *con-dieig* ‘PV-see.3S’ is the finite counterpart of the verbal noun *cuindchid*.

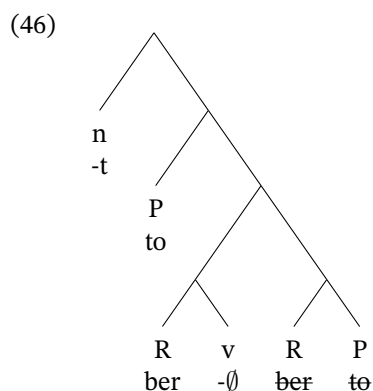
28. Verbal nouns can be contained within larger structures which raise to the clausal periphery as topics, but these cases do not affect the general patterns under discussion here.

29. Thurneysen (1946: 445) says that they are used “in place of the infinitive and (in combination with a preposition) of all participles other than the past participle passive”. He adds that “syntactically the verbal noun is a substantive” (p. 445).

- (45) *as-bert* *Conchubar* *fria*<sup>L</sup> *muintir* *ara* *scortis*  
PV-carry-3s Conchubar to.his retinue.ACC in.order.that unyoke.SUBJ.3p  
*a*<sup>N</sup> *gairptiu* *ocus* *ara* *cortis* *cor* *doL* *cuin-dchid*  
their chariots and in.order.that put.SUBJ.3p delegation to PV-seeking  
*tíge* *dóib*  
shelter.GEN for.them
- ‘Conchubar said to his party that they should unyoke their chariots and that they should put a delegation to the seeking of shelter for them.’  
(*Compert Con Culainn ‘The Conception of Cu Chulainn’, Part of the Ulster Cycle* 2023)

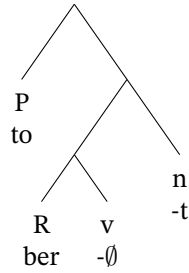
As this one verbal noun illustrates, the semantics of derived verbal nouns are typically predictable from the meaning of the matching verb. It follows that the same base structures should be found connecting the root and the preverb in both verbo-nominal and verbal clusters.

Transposing to OI verbo-nominal constructions the analysis that Branigan (2023) develops for similar constructions in Russian and Innu-aimûn, we propose that (i) preverbs originate as particles within the verb phrase (Svenonius 2005), so that the semantic unity of the verb and its prefix reduces to the familiar case of Germanic verb-particle pairings, that (ii) the particles adjoin to vP and that (iii) a nominalising higher n category triggers (multiple) head movement of the particle and of the verb, in that order. The derivation of the OI verbal noun *tabart* ([tavart]), related to *dobeir* ‘give, bring’, involves the syntactic structure (46), where the preverb originates as a particle complement to the verb and then adjoins to vP.



The lexical meaning of the verb and of the derived nominal comes from the combination of the root and of the particle in their base position. The surface morphological structure arises when the nominalising suffix attracts *do* and the verb in turn, with the latter tucking in beneath the particle. The structure (47) is the result.

(47)



From this structure, the appropriate cyclic phonological rules will generate the surface form *tabart*. Primary stress is assigned to the initial syllable in Irish; it falls on *to* in the structure (47). Lenition of the *b* onset to [v] in the second syllable—not reflected in the orthography—takes place immediately after the initial stressed vowel.

This analysis of “compound” verbal nouns relies on two premises. First, particles can adjoin to the verb phrase in the syntax. This might be taken as a particular instance of a very familiar syntactic operation which is already well-established in the literature: PP extraposition. PP extraposition seems to right-adjoin a PP to vP, so it is not immediately obvious that it is the same thing as adjunction of a particle.<sup>30</sup> But we assume that the surface ordering of adjunction structures is actually a post-syntactic word order decision and not the direct representation of any syntactic linear relationship. The structure (46) could equally be presented with the particle on the right. In that sense, the adjunction step in the derivation of (47) is hardly objectionable.

The other premise, noted already, is that parametric variation makes it possible for languages to identify particular heads as triggers for multiple head movement. Recall that the theory of head movement and of word formation adopted here rely on principle (39) and parameter (40). OI, like Russian or Innu-aimûn, is a language that resorts to MHM. The nominalising head in verbo-nominal constructions is one of the heads which has the relevant timing parameter set to allow MHM: Valuation of [u-stem] precedes labeling.

### 3.1.3 “Compound” verbs following a complementizer or a conjunct particle

Preverbs may appear with a conjunct verb to the right of a complementiser, as in (48).

- (48) a. ... *are nind-arbe analchi óod*  
           that PV-banish.3s.Subj vices from.him  
           ‘...that he banish from him his vices.’      Thes. 2, 244.21; Adger (2006: 612)
- b. *ní di-lg-aid a<sup>N</sup>-ancride do<sup>L</sup>-gní-ther fri-b act*  
    NEG PV-forgive-2p the.injury PV-REL-make.3s-PASS towards.2p but  
    *a-t<sup>L</sup>-gair-ith*  
    PV-3s-complain-2p

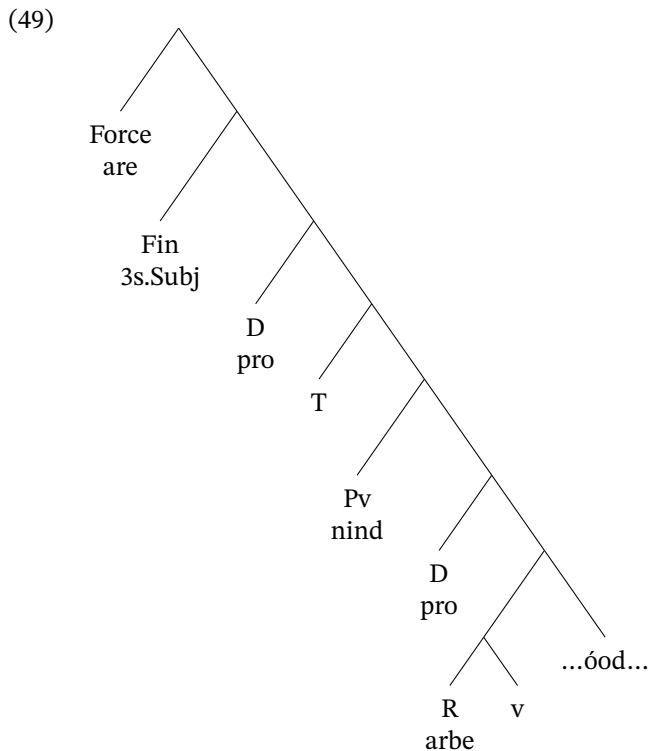
30. PP extraposition is often associated with non-syntactic factors, such as the prosodic weight of the extraposed PP. We take these to be an indication that adjunction operations can serve purposes beyond what the narrow syntax motivates. In the Old Irish structures, the particle adjunction operations are evidently motivated by the need to find a place for the preverb within a larger word. See Branigan (2023) for analysis of comparable cases in other languages.

‘you forgive not the injury that is done to you, but you complain about it.’  
 (García-Castillero 2020: 343)

- c. *ní ta-bir*  
 NEG PV-bring.3s

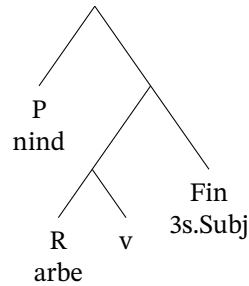
‘S/he does not bring.’ (McCone 2005: 72)

The challenge posed by these structures is how the preverb—which originates within the verb phrase—finds its surface position within the left periphery. In the verbo-nominal structures discussed above, the answer we proposed was that the verbo-nominal suffix triggers the movement of the preverb adjoined to vP and then the movement of the verb. The same approach accounts readily for the distribution of preverbs in conjunct finite structures. We need only suppose that Fin has the same ability to trigger MHM as the verbal noun suffix. Take (48a), for example, where the subject agreement features are expressed in Fin. In that case, the structure of (48a) will be (49), before any head-movement takes place at the clause level. (The preverb *nind* is already adjoined above vP at this point, as discussed above).



The inflection in Fin then attracts *nind* and then *arbe* upwards, forming the structure (50). The result is an  $X^0$  structure in which a “tucking in” ordering is found, so that the verbal stem ends up more closely attached to the inflection in Fin than the preverb *nind* is.

(50)



Primary stress is then assigned to the preverb within the full verb inflected with Fin, and the usual phonological effects accrue.

This survey of the syntax of preverbs in OI will be complete only when deuterotonic conjunct structures with a preverb in absolute initial position will be considered. This is done in the next section.

### 3.1.4 Preverbs in absolute initial position: deuterotonic conjunct structures

In each of the conjunct verb structures considered up to now, there is visible material to the left of Fin, which can be a complementiser or a negative particle or an interrogative pronoun. Preverbs, which originate lower in the clausal tree, are also attracted to Fin, together with the verb, in this context. And the internal morphophonology of the derived verbal form in Fin is largely independent of what appears to its left. But one also finds the conjunct inflection on the verbal form when a preverb stands at the left of the verb in Fin and is not preceded by higher material. In this case, the *deuterotonic* stress pattern obtains, and the expected phonological relationship between the preverb and the following verb disappears. Compare the forms in (51).

(51) a. *Ní epur a n-anman sund.*

*ní as-biur a n-anman sund.*  
NEG PV-say their names here

‘I do not say their names here.’

b. *as biur in-so*

PV say.1s this

‘I say this.’

In (51a), where the negative complementiser is present, the stress falls on the preverb *as*. The stressed vowel /a/ raises under the influence of the following high vowel and the derived consonant cluster /sb/ is reduced to /p/. In (51b), though, stress falls on the nucleus of the verb stem, the vowel in the preverb *as* remains unchanged and no cluster simplification takes place. This is the general pattern, as seen in examples (52).

- (52) a. *Ad cí ind rígain Conaire.*  
 PV see.3s the queen Conaire  
 ‘The queen sees Conaire.’ (Quin 1975)<sup>31</sup>
- b. *do moiniur*  
 PV think.1s  
 ‘I think’ (Thurneysen 1946: 27)
- c. *cita biat*  
 PV perceive.3p  
 ‘they perceive’
- d. *for con-gur*  
 PV PV-command.1s  
 ‘I command’

Phonological evidence indicates that the initial preverb is not contained within Fin in these cases, since it immediately precedes the deuterotonic juncture. This presents us with a significant challenge. The MHM process which normally combines the verb with its preverbs is sensitive to the core syntactic structure and should result in a configuration in which every preverb ends up in Fin, the highest one included, regardless of what the structure of the higher left periphery might be. This is not what happens in deuterotonic structures, however. For the correct configuration to be derived, it must be assumed that the leftmost preverb raises from the edge of vP to the position preceding Fin in the core syntax and that lower preverbs within the same clause can be attracted to Fin only when the highest one has raised up to Force.<sup>32</sup> Given our model of head-movement, this is unproblematic as long as the movement of the leftmost preverb precedes [u-stem] valuation, because only the uppermost copy of a chain is typically visible as a goal in valuation operations.

But the question remains why the leftmost preverb must raise. It is time to formulate the hypothesis that underlies our treatment of deuterotonic structures. We concur with [Carnie, Harley & Pyatt \(2000\)](#), who maintain that the initial preverb in a deuterotonic structure has raised to fill the C position. But in our view, the reason why raising to C, that is to Force, is necessary in this case is that the presence of material at the Force level prevents the deletion of the Force phase head. We propose therefore that preverb raising in the OI deuterotonic structures is forced precisely by the need to prevent deletion of the Force phase head. Technically, the preverb directly raises from the edge of vP to Force in the relevant structures and, once this syntactic event has taken place, Force cannot delete and retains its status as the clausal phase head. This movement patterns with  $\bar{A}$ -bar

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31. The example is taken from [Newton \(2006: 36\)](#), who attributes it to Quin (1975).

32. The details of similar derivations in other languages are presented in [Branigan \(2023\)](#).



movement, rather than with head-movement.<sup>33</sup> As such, it removes the initial preverb from the domain in which multiple head-movement applies.<sup>34</sup>

### 3.2 Adjoining preverbs to Force

To provide this proposal with a principled formal foundation, a few technical questions have to be articulated and answered. One of the most puzzling features of OI verbal complexes is that the first preverb in a sequence, like other prefixes in the language, sometimes act as a part of the verbal complex, other times as a separate  $X^0$  constituent, although the lexical meaning remains the same. In the latter case, it targets Force. Fin in turn hosts what remains of the verbal cluster, the second term in a sequence of preverbs and the finite verb itself. Once again, finite verbs in the corresponding structures appear to occupy a low position in the periphery, which we take to be Fin. Nominal subjects being hosted by the specifier of TP, the finite verb that precedes them cannot be realised in a high position in the inflectional domain, namely T, but is necessarily hosted by the lower head in the periphery.

The MHM model provides the resources to explain why the Old Irish preverbs can sometimes act as parts of the verb complex and other times as separate  $X^0$  constituents. Let us reconsider the role of phases in the divide between the absolute and conjunct inflections. In section 2, the fact that preverbs and absolute inflection are mutually exclusive was shown to follow from the way the richness of  $\varphi$ -features for labeling is computed in the derivations where Fin is a phase head. This account has a consequence that could potentially limit the expressive power of OI verb clusters. We have shown that the distribution of absolute verbs is syntactically predictable if the Force head must delete once it has transferred all of its formal features. But if it becomes a phase head, Fin cannot tolerate MHM. The implication is that simple root declarative clauses should not be allowed to use a preverb and a verb together. This is empirically false, of course, given the existence of examples like (31a), but the implication is still valid. The paradox is avoided in OI because the grammar has access to a *last resort* rescue operation which allows a Force head to escape deletion, even after it has transferred its features to Fin.

While deletion of a phase head is an obligatory operation when its conditions are met, it can only take place if deletion does not violate “recoverability”. Visible complementisers cannot be freely deleted. This constraint on deletion provides a loophole which OI makes use of.

When a preverb is found in a context where Force deletion should occur, the preverb is displaced to the edge of the phase to keep deletion from taking place. No feature checking or valuation is involved in this operation, which is driven entirely by labeling imperatives (we assume the derived position of the preverb involves adjunction, although the available OI data is perhaps inconclusive on this point). This movement is not driven by a

33. The nature of the complex entity resulting from preverb raising is not easy to characterize. We know that in deuterotonic structures, the juncture prevents any phonosyntactic process from taking place between the raised preverb and what follows the juncture. This suggests that two distinct phonological words are involved in this case. This leaves open the question of whether the verbal cluster as a whole defines a complex syntactic word or not. Dealing with verbal constructions in Middle Welsh and, to some extent, in Old Irish, Dedio (2023: 147) observes that the notion of word is not fully appropriate to deal with the patterns under consideration, because, if one sticks to standard definitions, it appears that they contain several domains that can be identified as morphological words and several domains that can be defined as phonological words.

34. The displaced preverb will still bear a [stem] feature, but since this feature is not involved in valuing a [u-stem] feature on Fin, constraint (39) does not require that the two be put together by the post-syntactic head-movement operation. The [stem] feature on the initial preverb can be interpreted in isolation in this case.

[u-stem] feature in Force, although the displaced preverb will itself bear a [stem] feature, as usual. This produces the *deuterotonic* form seen in (53).

- (53) *darmi regtais*  
 across go.COND.3p  
 ‘they would go over’ (Thurneysen 1946: 531)

### 3.2.1 Syntactic and phonological effects of adjunction

The adjunction of the preverb to Force has several effects. First, it ensures that Force cannot be deleted, because deletion of a category containing an adjunct would violate recoverability, even if the host for the adjunction were itself empty. Second, the addition of the content of the preverb itself will enrich the silent Force head sufficiently to allow the combination to serve as a label.<sup>35</sup> This means that deletion of Force is simply not an option in this context. The result is that the Force phase head will remain in place and that Fin will not become a derived phase head.

Adjunction of the preverb has effects on the phonology, too. Since *darmi* in (49a) and *ad* in (49b) are not contained within the word formed by head-movement to Fin, the usual word-level phonological effects will not apply to these preverbs. So the stress in (49a) will fall on the initial syllable of the verb *retais* and not on the preverb. The same is true in (49b), but here there are additional effects. Since *ad* and *gladathar* are never joined together, no consonant cluster simplification applies to the /d/ + /g/ string, unlike what occurs in the equivalent prototonic form *accaldathar*. And this result generalises to other differences between deuterotonic forms and their prototonic counterparts.

While the OI data do not provide transparent evidence concerning the specific landing site of the raised deuterotonic preverb, we find a formal parallel in the Icelandic Stylistic Fronting phenomenon Maling (1990), Holmberg (2000), Ingason & Wood (2017). Here too, an  $X^0$  category is seen to raise into the clausal left periphery to make up for the absence of something else, cf. (50).<sup>36</sup> The fronted category appears to the left of the finite verb, which optionally appears in second position in both root and embedded clauses in Icelandic Holmberg & Platzack (1995). And as in OI, the Icelandic movement can affect only the highest available category, as seen in (54), from Holmberg (2000).

- (54) a. *Hver sagðir þu að hefði sennilega skrifað þessa bók?*  
 who said you that has probably written this book  
 ‘Who did you say has probably written this book?’  
 b. *Hver sagðir þu að sennilega hefði skrifað þessa bók?*  
 who said you that probably has written this book

35. In order to be complete, the analysis should specify what is the feature that the raised preverb is endowed with which provides the whole clause with a label. We know that this feature is neither [negative], nor [interrogative], nor [topic], nor [focus], nor [relative]. We could decide to name the feature that is negatively specified for all the preceding ones [neutral declarative]. The question will be left open here.

36. Along similar lines, Rouveret (2024) argues that the verb-second constraint in German—the necessity of placing one expression, whatever it is, in the left periphery—should be traced back to the labeling requirement.

‘Who did you say has probably written this book?’

- c. *sá sem skrifað hefur (\*sennilega) þessa bók*  
 he that written has probably this book

The point here is not that OI and Icelandic both use syntactic fronting for the same purpose, or even that preverb fronting and Stylistic Fronting are the same type of formal operation. But the fact that Icelandic grammar sometimes resorts to the displacement of the highest available phonetically visible word to solve a structural problem means that UG must accommodate this option. In that case, the finding that Old Irish does the same thing should not cause concern.

The adjunction of a preverb to Force is also used in OI for other purpose, but with the same effect; it serves to block the deletion of Force. Verbs inflected in the imperfective, in the past subjunctive, or the conditional cannot bear absolute endings, so they must not become phase heads through Force deletion. When verbs with these endings would otherwise be initial within the left periphery, *no* insertion must take place to block Force deletion.<sup>37</sup> This is seen in examples (55), with *no* used to permit the expression of imperfective, past subjunctive, and conditional forms, in turn. In (55b), in fact, *no* accomplishes two functions at once, as it also provides a host for the *s* object clitic with a verb which agrees with a 1s subject.

- (55) a. *no scrútain*  
 PV investigate.IMPF.1s  
 ‘I used to investigate’ (McCone 1997: 86)
- b. *no=s bered*  
 PV=3 bear.PST.1s  
 ‘He was carrying it.’ (Thurneysen 1946: 261)

### 3.2.2 The nature of preverb movement

In order to conclude this section, it is important to be clear about the difference between the two types of preverb movement found in OI. The movement of the verb and of any number of preverbs to the Fin position is standard (multiple) head-movement, which takes place during the process of externalisation to form word-sized units which the morphophonology can process. This type of operation does not affect the core phrase structure. As such, head-movement to Fin does not alter the structural relationships which allow the *projection* of a label within an endocentric head-complement structure.<sup>38</sup> Note that this result is critically important for the labeling algorithm to succeed. Chomsky (2015) derives the *that*-trace effect by supposing that the syntactic objects which provide a label cannot be displaced until the phase that contains them is complete. His analysis is

37. Thurneysen (1946: 348, 370) notes that *no* is sometimes left out in poetic forms.

38. Working within an earlier model where  $X^0$  movement is always understood to be head-movement, Carnie, Harley & Pyatt (2000) propose that the fronting of the preverb is an instance of long head-movement, in the sense of Lema & Rivero (1989). Since  $X^0$  movement need not be conflated with head-movement in our model, preverb fronting instead patterns with other types of  $\bar{A}$ -movement, such as the *wh*-movement examined in section 2.7.2.

concerned primarily with the components of endocentric structures, but the logic should be the same for projecting heads. Syntactic movement of a true  $X^0$  head should always be impossible before the phase that contains it has been completed.

The preverbs which raise to the Force phase head also are uncontroversial  $X^0$  heads, but their movement is not a head movement in the standard sense. They raise from a position at the edge of the verb phrase to adjoin to a higher head position. But in doing so, they are displaced past some higher true syntactic heads, including Fin. We know that they move to resolve a problem which cannot be solved more locally, precisely because the higher heads cannot undergo syntactic movement themselves. Because preverb fronting is an operation that takes place for labeling purposes and is thus relevant to semantic interpretation, it is a movement within the core syntax, which must have taken place at Transfer at the latest. It removes the preverb from the search path of the Fin probe. As such, when Fin probes one or more goals to value its [u-stem] features, the preverb raised to protect Force from deletion will not be available. The highest preverb thus does not qualify as an intervener for post-syntactic head-movement.

When there are several preverbs present, it is always the outermost one which moves to the phase edge. All remaining preverbs are subject to MHM and target Fin, where the expected prototonic phonology is found. In (56), for example, the preverb *to* raises to the phase edge to block Force deletion and the preverb *ind* is attracted to Fin with MHM. The latter is assigned initial stress within the Fin word and undergoes cluster reduction within that same domain.

- (56) a. *do ind-so-am* [to-ind-soam ‘turn, translate’]  
           PV PV turn-1p  
           ‘we turn, translate’ (Thurneysen 1946: 519)

## 4 Conclusions

We have shown that the major complexities in OI verbal syntax and morphology can be understood as the consequences of quite general features of the grammar. The first one is a clause structure in which the left periphery of sentences is articulated in at least two distinct heads, Force and Fin, and where Fin often triggers multiple head-movement. The second one is the major role played by the circulation of features in the left periphery, in particular via the Feature Inheritance process, involving Force and Fin, which directly contributes to the identification of active phase heads. The third one is precisely the claim that derivations proceed by phase, which is of crucial importance for the availability and survival of features in derivations. Finally, the labeling requirement, combined with phase theory, appears to be an essential dimension in the formation of verbal clusters and in the divide between absolute and conjunct verbal forms. The fundamental observation here is that a verbal form inflected for conjunct morphology cannot it and of itself provide a label for the whole clause. Sometimes the verb stands in initial position, but in this case, this position must be identifiable as a phase head and the verb must be marked for absolute inflection. At other times, the verb is not in initial position, it is marked for the conjunct morphology and the label is provided by the initial element, which can be a conjunct particle, a complementiser or a dissociated preverb. We end up with a syntactic view of verbal

clusters, such that the analysis of all their relevant characteristics keeps almost entirely within the bounds of the syntactic component. These include the internal make-up of clusters, the nature of the initial position, the site occupied by verbal heads, the absolute/conjunct asymmetry, the distribution of clitics, and the position of the deuterotonic juncture. In our approach, the only feature of OI verbal clusters that requires a reference to the “post-syntax” component is the claim that head movement involves two components, a [u-stem] feature valuation operation which takes place within the core syntax at the phase level and a substitution operation which operates as part of the externalization process, forming complex words from atomic roots and affixes.

Several aspects of the analysis have implications that extend beyond Celtic to broader theoretical questions. As just mentioned, the OI materials provide a new source of information concerning the relationship between verbal inflection and the labeling algorithm. Successful endocentric labeling requires a transparent connection within *phase level* syntax and the externalised output of morphological inflection. This result has implications for the general architecture of the derivation. At the very least, it means that the computation must have access to some of the morphological features of the terminal nodes within phase level syntax, contrary to what strong versions of the *late insertion* hypothesis (Marantz 1995, Harley 2014) would imply. More generally, it affirms the importance of paying attention to the intricate morphophonological patterning in languages like OI, where the payoff is not only for morphological theory, but also for our understanding of core syntax.

The proposed analysis has other far-reaching implications. Innovations in grammatical theory must initially be judged by the arguments used to motivate them, but their value becomes more apparent if they provide new tools for understanding phenomena which initially appear to be quite unconnected. Chomsky’s (2015) proposal that phase head status can be transferred when the original phase head is deleted is put forward as a part of an explanation for the *that*-trace effect. The distribution of absolute and deuterotonic verb forms in OI has no obvious connection to constraints on *wh*-movement in English, and yet we have shown here that it is the same underlying phenomenon which is responsible for both, if the analyses provided here are on the right track. Along the same lines, the model of multiple head-movement in Branigan (2023) was constructed to explain the derivational morphology of Algonquian languages and the use of aspectual prefixes in Slavic. Neither of these are immediately related to the VSO word order and the morphophonology of complex verbs in OI. And yet a hallmark of OI syntax seems also to be that multiple head-movement derivations are commonplace.

## Textual Abbreviations

CCC *Compert Con Culainn*. A. G. van Hamel(ed). Dublin Institute for Advanced Studies, 1978.

Ir. Gosp. The Irish Gospel of Thomas.

LL The Book of Leinster, formerly Lebar na Núachongbála. R. Best, M. O’Brien & A. O’Sullivan (eds.) (1954–1983). Dublin: Dublin Institute for Advanced Studies.

LU Lebor na hUidre (Book of the Dun Cow). Best R. & O. Bergin (eds.)(1929). Dublin: Dublin Institute for Advanced Studies.

Ml The Milan glosses on the psalms, Milan. Thes i, 7–483.

TBF T'ain B'o Fro'ich

Thes. Thesaurus Palaeohibernicus: a collection of old-Irish glosses, scholia, prose, and verse. W. Stokes & J. Strachan (eds.) (1901–3) Dublin Institute for Advanced Studies.

Wb Würzburg Glosses on the Pauline epistles. Thes i, 499–712.

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